Hints: Probability – 4. The probabilistic method across mathematics

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Theorem 8: if $X < c$ at every outcome then $\mathrm{E}X < c$; if no good outcome existed, $\Pr(\text{bad}) = 1$. Alteration: if a random object has $\mathrm{E}(\text{size} - \#\text{bad parts}) \geqslant s$, some outcome achieves this, and deleting one element from each bad part leaves a good object of size $\geqslant s$.

Hint 2. Ramsey: colour the edges of K_n independently red/blue. The expected number of monochromatic K_k is $\binom{n}{k}2^{1-\binom{k}{2}}$; for $n = \lfloor 2^{k/2} \rfloor$ and $k \geqslant 3$ this is < 1 (use $\binom{n}{k} < n^k/k!$). So some colouring has none, and $R(k,k) > n$.

Hint 3. Bipartite subgraph: put each vertex independently in one of two parts. Each edge is cut with probability $\frac{1}{2}$, so the expected number of cut edges is $e/2$, and some partition cuts at least $e/2$ edges.

Problem 2

Hint 1. Bound the number of regions C (components of the graph G on the squares, joining adjacent squares of equal colour) below by a quantity whose expectation is easy. A forest bound $C \geqslant V - E$ is not enough.

Hint 2. G is planar. By Euler's formula $V - E + F = 1 + C$, so $C = V - E + (F - 1) \geqslant V - E + Q$, where Q is the number of monochromatic 2×2 blocks (each bounds its own face).

Hint 3. Take expectations: $V = mn$, $\mathrm{E}E = \frac{1}{2}(m(n-1) + n(m-1))$, $\mathrm{E}Q = \frac{1}{8}(m-1)(n-1)$. You get $\mathrm{E}C \geqslant \frac{mn+3m+3n+1}{8} > \frac{mn}{8}$.

Problem 3

Hint 1. What do all nilpotent matrices have in common that is a linear condition?

Hint 2. A nilpotent matrix has all eigenvalues 0, hence trace 0, and the trace is linear, so the span lies in the proper subspace $\{\mathrm{tr} = 0\}$. Probabilistic form: a random linear combination of nilpotent matrices has trace 0 with probability one, whereas a random matrix of $M_n(\mathbb{F})$ (e.g. Gaussian over $\mathbb{R}$, uniform over a finite field) has non-zero trace with positive probability; so the span is not everything.

Problem 4

(a)

Hint 1. All entries are $\geqslant 1$, so A is never scalar. For a non-scalar 2×2 matrix A, the matrices commuting with it are exactly $\alpha A + \beta I$.

Hint 2. Write $B = \alpha A + \beta I$. Given A (N^4 choices) and the entry $f = B_{12}$ (N choices), $\alpha = f/A_{12}$ is fixed, which determines B_{21} and the difference $B_{11} - B_{22}$; choosing B_{11} (N choices) determines B. Hence at most N^6 commuting pairs.

(b)

Hint 1. Divide by the N^8 equally likely pairs: $\Pr(AB = BA) \leqslant N^{-2} \to 0$.

Problem 5

Hint 1. $M = A - A^T$ is skew-symmetric with independent entries $x_{ij} = a_{ij} - a_{ji}$ $(i < j)$, each 0 w.p. $\frac{1}{2}$ and ± 1 w.p. $\frac{1}{4}$; so $\mathrm{E}x_{ij} = 0$, $\mathrm{E}x_{ij}^2 = \frac{1}{2}$. Expand $\det M = \sum_\sigma \operatorname{sgn}\sigma \prod_i M_{i\sigma(i)}$ and take expectations term by term.

Hint 2. A term has non-zero expectation only if every variable occurs an even number of times. Since $M_{ii} = 0$ and x_{ij} occurs only via M_{ij}, M_{ji}, this forces σ to be a fixed-point-free involution; then the term is $(-1)^n \prod(-x_{ij}^2) = \prod x_{ij}^2$, with expectation 2^{-n}.

Hint 3. There are $(2n-1)!!$ such involutions, so $\mathrm{E}\det(A - A^T) = \frac{(2n-1)!!}{2^n} = \frac{(2n)!}{4^n n!}$.